THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS UNIT TEST SET C

Chapters 8–11: **Introduction to Trigonometry, Some Applications of Trigonometry, Circles, Areas Related to Circles** | RBSE Board Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat figures; use graph paper where a graph is asked.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. The value of $\frac{2\tan 30^\circ}{1-\tan^2 30^\circ}$ is: 1

(a) 0 **(b)** $\frac{1}{\sqrt{3}}$ **(c)** $\sqrt{3}$ **(d)** 1

2. If $\sin A = \frac{5}{13}$, then $\cos A$ is: 1

(a) $\frac{12}{13}$ **(b)** $\frac{8}{13}$ **(c)** $\frac{5}{12}$ **(d)** $\frac{13}{12}$

3. $\frac{1-\tan^2 45^\circ}{1+\tan^2 45^\circ}$ equals: 1

(a) $\sin 45^\circ$ **(b)** 0 **(c)** $\tan 90^\circ$ **(d)** 1

4. A kite is flying at a height of 60 m; its string makes 60° with the ground. The length of the string is: 1

(a) $30\sqrt{3}$ m **(b)** $40\sqrt{3}$ m **(c)** 120 m **(d)** $60\sqrt{3}$ m

5. A 6 m high pole casts a shadow $2\sqrt{3}$ m long. The sun's elevation is: 1

(a) 60° **(b)** 30° **(c)** 45° **(d)** 90°

6. An observer 1.5 m tall is 28.5 m from a chimney; the angle of elevation of its top from his eyes is 45°. Student A says the chimney is 28.5 m tall; Student B says 30 m. Then: 1

(a) Student A is correct **(b)** Student B is correct
(c) Both are correct **(d)** Both are wrong

7. From a point 10 cm away from the centre, the length of the tangent to a circle of radius 6 cm is: 1

(a) 6 cm **(b)** 8 cm **(c)** 16 cm **(d)** 4 cm

8. The length of the tangent from a point A 5 cm from the centre is 4 cm. The radius is: 1

(a) 3 cm **(b)** 5 cm **(c)** $\sqrt{41}$ cm **(d)** 4 cm

9. A quadrilateral ABCD circumscribes a circle with $AB = 8$, $BC = 9$, $CD = 7$ cm. Student A says $AD = 6$ cm; Student B says $AD = 8$ cm. Then: 1

(a) Student A is correct **(b)** Student B is correct
(c) Both are correct **(d)** Both are wrong

10. The area of a sector of a circle of radius 14 cm and central angle 45° is: 1

(a) 77 cm² **(b)** 38.5 cm² **(c)** 308 cm² **(d)** 154 cm²

11. A student states: "If the radius of a sector is doubled keeping the angle same, its area doubles." The correct response is: 1

(a) Correct **(b)** Incorrect — the area becomes 4 times
(c) The area is unchanged **(d)** The area halves

12. The length of an arc subtending 45° in a circle of radius 28 cm is: 1

(a) 22 cm **(b)** 44 cm **(c)** 88 cm **(d)** 11 cm

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

13. A student claims that $\sin(A+B) = \sin A + \sin B$. Check the claim for $A = B = 30^\circ$. 2

14. A student says that the angle of elevation of the top of a 20 m tower from a point 20 m from its foot is 60°. Justify or refute. 2

15. AB is a chord of a circle with centre O and AT is the tangent at A such that $\angle BAT = 60^\circ$. Find $\angle AOB$. 2

16. PA and PB are tangents to a circle of radius 5 cm with $\angle APB = 90^\circ$. A student says $PA = 5\sqrt{2}$ cm. Justify or refute. 2

17. The length of the minute hand of a clock is 14 cm. Find the area swept by it in 5 minutes. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

18. In $\triangle PQR$, right-angled at Q, $PR + QR = 25$ cm and $PQ = 5$ cm. Determine $\sin P$, $\cos P$ and $\tan P$. 3

19. A 1.6 m tall statue stands on the top of a pedestal. From a point on the ground, the angle of elevation of the top of the statue is 60° and that of the top of the pedestal is 45°. Find the height of the pedestal. 3

20. From an external point P, tangents PA and PB are drawn to a circle with centre O. If $PA = 12$ cm and $\angle APB = 60^\circ$, find the length of chord AB and $\angle OAB$. 3

21. An arc of a circle of radius 35 cm subtends 72° at the centre. Find 3
(i) the length of the arc
(ii) the area of the sector
(iii) the perimeter of the sector.

22. Prove that the tangent at any point of a circle is perpendicular to the radius through the point of contact. 3

OR

Prove that the tangents drawn at the ends of a chord of a circle make equal angles with the chord. 3

23. Prove that $\frac{1+\tan^2 A}{1+\cot^2 A} = \left(\frac{1-\tan A}{1-\cot A}\right)^2 = \tan^2 A$. 3

OR

Express $\sin A$, $\sec A$ and $\tan A$ in terms of $\cot A$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

24. **Case study:** A window of a house is 10 m above the ground. From the window, the angles of elevation and depression of the top and bottom of another house on the opposite side of the lane are 60° and 45° respectively. 5
(i) Draw a figure.
(ii) Find the width of the lane.
(iii) Find the height of the other house.

OR

Case study: Two men on either side of a cliff 80 m high observe the angles of elevation of the top of the cliff to be 30° and 60° respectively. 5
(i) Draw a labelled figure.
(ii) Find the distance of each man from the foot of the cliff.
(iii) Find the distance between the two men.
(iv) Which man is nearer the cliff and why?

25. **Case study:** A circular cake of radius 21 cm is cut into 6 equal sector-shaped pieces. 5
(i) Find the angle of each piece.
(ii) Find the area of the top of each piece.
(iii) Find the length of the curved crust of each piece.
(iv) A student says 2 pieces together cover 462 cm². Verify.

OR

Case study: A hand fan is in the shape of a sector of radius 28 cm and angle 135°. 5
(i) Find the area of the fan.
(ii) Find the length of lace needed along its curved edge.
(iii) Find the total cost if lace costs ₹2 per cm and cloth costs ₹0.50 per cm².

END OF QUESTION PAPER — **Total: 50 Marks**